EXNO: 7

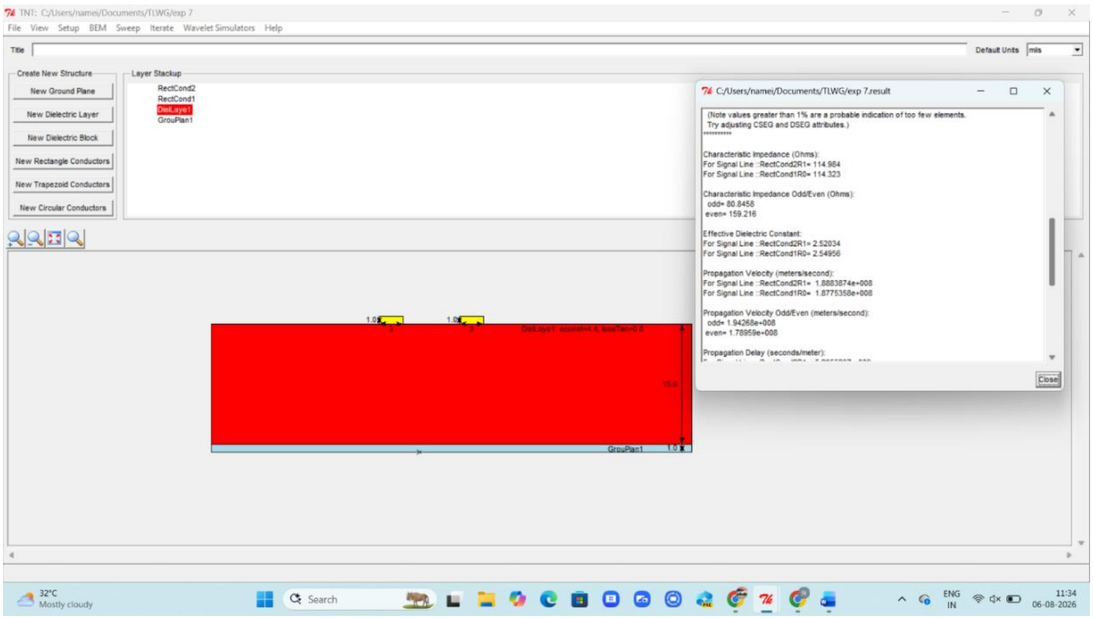
Design a two conductor transmission line and Investigate the impact of Dielectric layer thickness on characteristic Impedance and Propagation Velocity

Test Case 1: Design a Two conductor transmission line with single ground plane and estimate the impact of varying the thickness of the dielectric layer on characteristic Impedance and Propagation Velocity. Thickness to be varied between 5 to 15mm.

OUTPUT:

Thickness (mm)	**Simulation (Z_0)**	**Simulation Velocity**	**Theoretical (Z_0)**	**Theoretical Velocity**
5	80.38 Ω	(1.816\times10^8) m/s	**80.00 Ω**	**(1.82\times10^8) m/s**
10	102.61 Ω	(1.863\times10^8) m/s	**102.00 Ω**	**(1.86\times10^8) m/s**
12	108.142 Ω	(1.872\times10^8) m/s	**108.00 Ω**	**(1.87\times10^8) m/s**
15	114.65 Ω	(1.882\times10^8) m/s	**114.00 Ω**	**(1.88\times10^8) m/s**

Simulation Results:



Test Case 2: Design a Two conductor transmission line with single ground plane and investigate the effect of far end cross talk and near end cross talk by varying the thickness of the dielectric layer .Thickness to be varied between 5 to 15mm.

OUTPUT:

Thickness (mm)	**Simulation FEXT (dB)**	**Simulation NEXT (dB)**	**Theoretical FEXT (dB)**	**Theoretical NEXT (dB)**
5	-44.06	-34.57	**-44.00**	**-34.50**
10	-44.63	-29.25	**-44.50**	**-29.20**
12	-45.15	-28.25	**-45.00**	**-28.20**
15	-45.90	-27.11	**-45.80**	**-27.10**

Simulation Results: